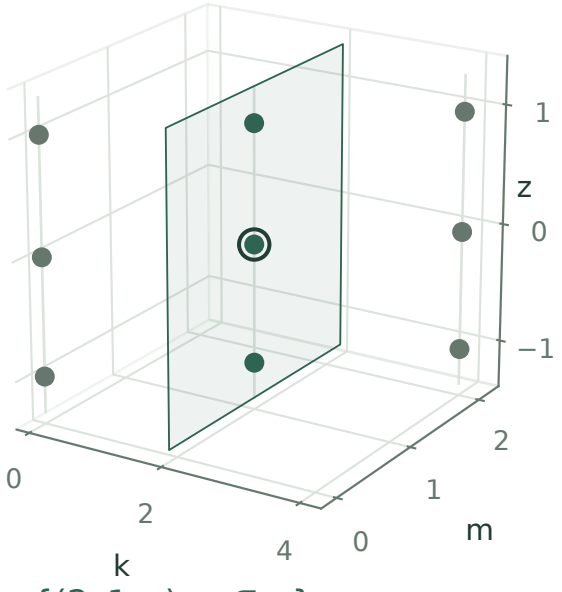
When does the prescribed slice meet the lattice?

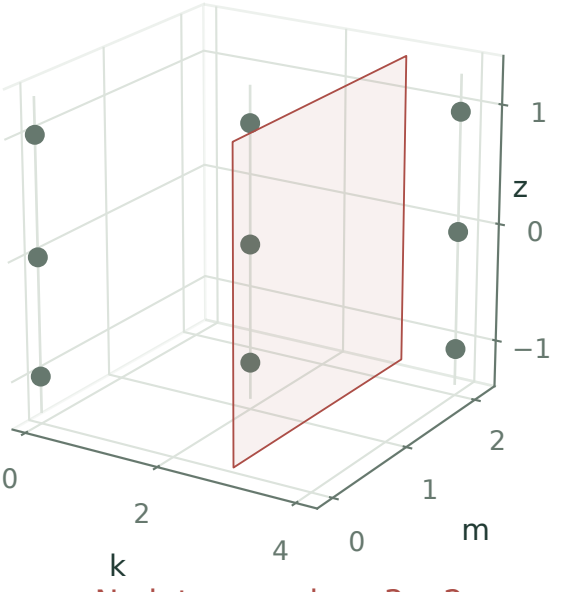
Exact families have integer coordinates and satisfy k = 2m.

k = 2: feasible



$\{(2, 1, z) : z \in \mathbb{Z}\}$

k = 3: blocked



No integer solves $3 = 2m$.